

Aaditya Sakhardande

+1 (480) 791-9046 | aadityasakhardande@gmail.com | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

EDUCATION

Arizona State University, Tempe, AZ, USA

May 2026

Masters in Robotics and Autonomous Systems (Electrical Engineering)

3.7 GPA

Mukesh Patel School of Technology Management and Engineering, Mumbai, India

August 2024

B.Tech. Major in Electronics and Telecommunication

3.11 GPA

WORK EXPERIENCE

GrayMatter Robotics, Los Angeles, CA – Robotics Engineer – Systems and Applications

August 2026 - Present

- Intensive process development for on site Robot deployment at customer sites, including smooth installation for operational hand-off.

GrayMatter Robotics, Los Angeles, CA – Robotics Engineer

January 2026 – May 2026

- Programmed and deployed FANUC robots for automated scanning and sanding of complex production parts, including on-site system activation and integration at customer facilities.
- Integrated LMI, Zivid, Alvium vision systems to enable 3D scanning and adaptive robot motion for irregular geometries.
- Developed ROS2-based robotic applications and managed Docker systems for scalable, reliable deployment across development and production environments.
- Extended robotic sanding workflows with error detection mechanisms for over-sanding and under-sanding, integrating feedback for improved precision on complex geometries.

Leyline Technologies, Spokane, WA - Product Engineer

June 2025 – August 2025

- Tasked to design and document 3 AI models that optimize product features in alignment with real-time robotics systems.
- Iterating 2 data-driven prototypes, leveraging Python and internal tools to improve model accuracy and pipeline clarity by 20%.
- Collaboration with cross-functional teams to integrate product requirements into modular designs for robot-adjacent use cases.

Cartken | Arizona State University, Tempe, AZ – Prototype Team Lead

May 2025 – August 2025

- Designed and prototyped robotic subsystems using reclaimed components, focusing on hardware and embedded control.
- Analyzed robotic platforms' environmental impact and hardware design, sustainable manufacturing, and reuse strategies.

MaitriAI, Mumbai, IN – Software Developer

December 2023 - May 2024

- Trained CNN model reaching 98% accuracy for document identification. Enhanced Biometric OCR Software, extracting text with 94% precision from scanned documents. Implemented SQL database to store extracted documents and texts.
- Conceptualized enterprise architecture of LMS platform, enabling creation and management of training programs.
- Built and deployed 12 APIs employing Python libraries such as PyTesseract, TensorFlow, and FlaskAPI.

MPSTME University, Mumbai, IN – Student Researcher

August 2023 – November 2023

- Led **Online Canteen Ordering Management System** project, building a web app for online orders, customizable menus, and tracking 300+ order statuses in real time.
- Leveraged and maintained system using **HTML, CSS, JavaScript, PHP, MySQL, and XAMPP server**.
- Coordinated UI design, integrated payment gateway, and optimized SQL schema for transactions and metadata storage.

RESEARCH PUBLICATIONS

Robotic Gait Trainer with Exoskeleton ([Link to the Paper](#))

February 2026

- Headed research on robotic gait trainer for spinal cord injury patients, conducting a 25-year trend analysis of functional models.
- Analyzed rehabilitation practices and BCI strategies to develop six optimal models for neural function recovery.
- Presented at the *International Conference on Materials, Robotics, Automation, Computer and Control (ICMRACC 2025)*.

PROJECTS

Drone Vision Navigation Framework (Project)

Spring 2025 – Summer 2025

- Engineered a real-time vision system for the Parrot Mambo UAV drone using MATLAB/Simulink, achieving 92.3% landing accuracy across 50+ test flights via closed-loop feedback control.
- Used HSV color segmentation and centroid localization on a 640x480 video feed at 30 fps to extract spatial features in real-time.
- Designed control laws using normalized image-plane error, cutting lateral drift by over 83% and improving convergence speed.

Real-time Video Inpainting Software (Project)

Spring 2025

- Built an object removal pipeline using Mask R-CNN and OpenCV (96% mask accuracy, 30 fps), integrating GAN-based inpainting models (LaMa, Stable Diffusion) for texture-consistent restoration.
- Optimized for achieved 1,000 frames per minute, supporting videos up to 1080p resolution with minimal artifacts.

Autonomous Maze Solver using MyCobot Pro 600 (Project)

Fall 2024 – Spring 2025

- Engineered and built digital twin to simulate forward and inverse kinematics, ensuring precise and efficient navigation.
- Programmed MyCobot Pro 600 to autonomously solve a 4x4 maze, optimizing path planning and obstacle detection.
- Automated and executed a maze-solving algorithm incorporating Python, MATLAB, and SOLIDWORKS, merging socket programming and image processing for real-time maze recognition.

TECHNICAL SKILLS

Programming: Python, C++, JavaScript, Bash, Pandas, OpenCV, TensorFlow, PyTorch, SQL, PHP.

Software: MATLAB/Simulink, ROS2, Linux, Git, Anaconda, AutoCAD, SolidWorks, RepetierHost, Proteus, NerfStudio.

Other Skills: Project Management, GANs, Docker & REST APIs, Image Segmentation, PLCs, Control Systems (PID, Filter), NLP.

Certifications: 100 Days of Code (Python – Udemy), Advanced Python (LinkedIn), Learning C# (LinkedIn)